
LingYue Digital
CIS Delivery Standard

A revision history

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1. Introduction

1.1 Purpose

Standardize and clarify the delivery standards for LingYue Digital project, provide guidance and direction for the subcontractors to establish the delivery process and product quality, help each team to establish engineering quality standards and objectives, assist in the acceptance of relevant software systems, and improve the overall quality efficiency within the organization.

1.2 Scope

LingYue Digital implements research and development projects and outsourcing product projects.

2. Implement project quality requirements

Supplier development team should develop and maintain the integration of interfaces, databases, platforms and third-party systems of the required systems in accordance with The business requirements of LingYue; After the completion of coding, code walk - through, unit testing, etc.

The testing process and activities must comply with the quality system specifications of LingYue, and the process delivery and management must be carried out on Lingyue's tool platform (such as Confluence and JIRA).

The number and maturity of development and testing personnel should meet the standards required by LingYue. The work arrangements of the testers shall be incorporated into the quality management system of LingYue, and the work should be carried out in strict accordance with LingYue's quality standards. The work of the testers is assigned and managed by LingYue QM staff and make daily work reports.

Functional test access criteria:

(1)Requirements document review passed, Jira/Confluence review passed

requirements document can be accessed;

(2) The unit test is completed and the export conditions are met (the coverage rate of logic class shall not be less than 90%, the passing rate shall be 100%, the coverage rate of page class shall not be less than 85%, and the passing rate shall be 100%);

(3) Smoke test completed and exit conditions satisfied (pass rate > 95%);

Functional test criteria:

(1) The coverage of test cases to requirements reaches 100%;

(2) The execution rate of test cases reached 100%;

(3) The pass rate of test cases reaches 95%;

(4) All found defects are recorded and tracked;

(5) The defect repair rate reached 95%;

(6) No left Critical and High level defects;

(7) The test report was confirmed by the evaluation.

(8) Test Plan/Test Case/Test Execution/Defect/Test Report meets the requirements of JIRA LingYue Digital Test Management Rules.

Test process quality standard:

- ◆ The test should be carried out according to the test schedule;
 - ◆ Testers participate in requirements explanation and review, and fully understand the requirements content and system implementation mode;
 - ◆ Analyze test requirements, write test cases and cover all requirements., Test cases must be reviewed and confirmed by PO and development team;
 - ◆ Defects found in the test process shall be transferred on the JIRA platform, and the test defects should be submitted, tracked and managed until closed;
 - ◆ The test execution must meet the process quality management requirements of LingYue, such as SIT test, Pre production environment test, production smoking test, etc;
 - ◆ The communication during the test should comply with LingYue's
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management specifications;

- ◆ After the test, the test result report should be submitted according to the requirements of LingYue, which must be reviewed by the test leader in charge of LingYue and meet the filing requirements.

3. Quality requirements for outsourced products

For externally purchased products, the supplier should meet the functional and performance requirements proposed by LingYue, and provide relevant functional operation and system maintenance instructions; After the system is deployed, the supplier should be responsible for monitoring, troubleshooting and bug repair of the system platform according to the contract requirements, and the response process and standard should fully meet the SLA requirements of LingYue.

Product function quality standard:

- ◆ The system can start and exit normally;
- ◆ The product functions are fully consistent with the product requirements;
- ◆ The function of the product is stable, and there are no serious or fatal defects in the system operation;
- ◆ Browse all pages, connect correctly, and display all function buttons and interfaces correctly;

Non functional quality standard:

- ◆ Performance requirement: The key business and performance meet the requirements put forward by LingYue or meet the product performance requirements;
 - ◆ Stability requirements: Normal operation time/(normal operation time+failure time);
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- ◆ Robustness requirements: Mean time between failures/ year;

4. Evaluation indicators for suppliers

4.1 Project Implementation quality measurement assessment indicators and standards

The supplier's quality evaluation (test quality) will be comprehensively evaluated through the following indicators to form the final evaluation index item score.

1. Test quality evaluation and assessment

The test quality evaluation is carried out from two dimensions: one is the comprehensive index evaluation, which is obtained through the comprehensive evaluation of various indexes of the test; The second is special index evaluation, which evaluates important indexes.

The assessment of the project shall be carried out by the combination of comprehensive index evaluation and special index evaluation. If the indicator value exceeds a certain threshold, the appraisal results include the accumulation of comprehensive indicator appraisal results and single indicator appraisal results.

(1) Comprehensive index evaluation and assessment

NO.	Index name	Index definition	Assessment weight	Scoring criteria(Deduction of score items)
1	Defect leakage rate of SIT Test	(Pre-production defect+Defects found within 1 month after production)/(SIT defect+Pre-production defect+Defects found within 1 month after production)× 100%	20%	◆ Defect leakage rate>30%:deduct 100 points,And enter the link of special index evaluation and treatment ◆ 20%<Defect leakage rate≤30%:deduct 70 points ◆ 15%<Defect leakage rate≤20%:deduct 50 points ◆ 10%<Defect leakage rate≤15%:deduct

NO .	Index name	Index definition	Assess ment weight	Scoring criteria(Deduction of score items)
				20 points ◆ Defect leakage rate $\leq$ 10%:no deduction.
2	Pass rate of SIT test cases	(Number of pass use cases/Number of executed use cases) $\times$ 100%	15%	◆ Pass rate of test cases $\geq$ 95%:no deduction ◆ 95%>Pass rate of test cases $\geq$ 90% : deduct 20 points ◆ 90%>Pass rate of test cases $\geq$ 80% : deduct 60 points ◆ Pass rate of test cases<80%: deduct 100 points
3	SIT test case execution rate	(Number of executed use cases/Total number of test cases) $\times$ 100%	15%	◆ Test case execution rate $\geq$ 90%: no deduction ◆ 90%>Test case execution rate $\geq$ 80% : deduct 20 points ◆ 80%>Test case execution rate $\geq$ 70% : deduct 60 points ◆ Test case execution rate<70%: deduct 100 points
4	Proportion of SIT effective defects	(Effective defect number/The total number of defects) $\times$ 100%	20%	◆ Proportion of effective defects $\geq$ 95%: :no deduction ◆ 95%>Proportion of effective defects $\geq$ 90%: deduct 20 points ◆ 90%>Proportion of effective defects $\geq$ 85%: deduct 50 points ◆ 85%>Proportion of effective defects $\geq$ 75%: deduct 70 points ◆ Proportion of effective defects<75% : deduct 100 points,And enter the link of special index evaluation and treatment
5	JIRA operational compliance	JIRA platform usage standardization	20%	JIRA use normative problems found in monthly quality audi: ◆ Number of non-compliance issues<1-5: no deduction ◆ 5<Number of non-compliance issues $\leq$ 10: deduct 20 points ◆ 10<Number of non-compliance issues $\leq$ 20: deduct 50 points ◆ 20<Number of non-compliance issues $\leq$ 30: deduct 70 points

NO.	Index name	Index definition	Assessment weight	Scoring criteria(Deduction of score items)
				◆ Number of non-compliance issues>30And the problem has not been modified after three consecutive reviews,deduct 100 points,And enter the link of special index evaluation and treatment.
6	Process compliance	Problems found in quality audit	10%	Other process compliance problems found in monthly quality audit: ◆ Number of non-compliance issues<1-5: no deduction ◆ 5<Number of non-compliance issues≤10: deduct 20 points ◆ 10<Number of non-compliance issues≤15: deduct 50 points ◆ Number of non-compliance issues>15 And the problem has not been modified after three consecutive reviews:deduct 100 points;

The comprehensive evaluation score adopts the hundred mark system, which is obtained by accumulating the scores of each index item×weight:

Comprehensive evaluation score= "Defect leakage rate of SIT Test"score×weight+"Pass rate of SIT test cases"score× weight+"SIT test case execution rate"score× weight+"Proportion of SIT effective defects"score× weight+"JIRA operational compliance"score× weight+"Process compliance"score× weight.

For example, 20 points are deducted for "Defect leakage rate of SIT Test"score", 20 points are deducted for "Proportion of SIT effective defects", 50 points are deducted for "JIRA operational compliance", 50 points are deducted for "Process compliance" and no points are deducted for other items, so the comprehensive evaluation score is:

$$80 \times 20\% + 100 \times 15\% + 100 \times 15\% + 80 \times 20\% + 50 \times 20\% + 50 \times 10\% = 77 \text{ points.}$$

Compliance indicators, assessment standards and penalties:

	Target value	Index value	Punishment rules
Comprehensive evaluation score	$X \geq 90$ points	$85 \text{ points} \leq X < 90$ points	20% of the current quarter PO total amount will be deducted;
		$80 \text{ points} \leq X < 85$ points	50% of the current quarter PO total amount will be deducted;
		$X < 80$ points	100% of the current quarter PO total amount will be deducted;

(2) Special index evaluation and assessment

NO.	Index name	index value	Punishment rules
1	Defect leakage rate of SIT Test	$30\% < Y \leq 40\%$	10% of the current quarter PO total amount will be deducted;
2		$40\% < Y \leq 50\%$	25% of the current quarter PO total amount will be deducted;
3		$Y > 50\%$	100% of the current quarter PO total amount will be deducted;
4	Proportion of SIT effective defects	$70\% \leq Z < 80\%$	15% of the current quarter PO total amount will be deducted;
5		$60\% \leq Z < 70\%$	30% of the current quarter PO total amount will be deducted;
6		$Z < 60\%$	100% of the current quarter PO total amount will be deducted;
7	JIRA operational compliance [Number of questions\review times]	15-25 problems	20% of the current quarter PO total amount will be deducted;
8		25-40 problems	50% of the current quarter PO total amount will be deducted;
9		>40 problems	100% of the current quarter PO total amount will be deducted;
10		More than 3 problems have not been repaired by more than 3 rechecks	Meet with suppliers to discuss suspension of cooperation.

2. Development quality evaluation and assessment

N O .	Index name	target value	index value	Punishment rules
1	Smoke test pass rate [[Smoke Test Pass times/Total Smoke Test times) × 100%]	≥ 95%	90%(including)≤X<95 %	20% of the current quarter PO total amount will be deducted;
2			80%≤X<90%	50% of the current quarter PO total amount will be deducted;
3			X<80%	100% of the current quarter PO total amount will be deducted;
4	Defect repair rate [(Number of defects fixed / total number of effective defects) × 100%]	≥ 90%	80%≤Y<90%	20% of the total PO amount of this quarter will be deducted;
5			70%≤Y<80%	50% of the total PO amount of this quarter will be deducted;
6			Y<70%	100% of the total PO amount of this quarter will be deducted;
7	Proportion of serious defects [((Critical+High) number of defects/Total number of effective defects) × 100%]	≤20%	20%<Z≤25%	20% of the total PO amount of this quarter will be deducted;
8			25%<Z≤30%	50% of the total PO amount of this quarter will be deducted;
9			Z>30%	100% of the total PO amount of this quarter will be deducted;

4.2 Quality measurement index and assessment standard of outsourcing

N O.	Index type	Index name	target value	index value	penalty
1	Functional parameter	Functional aspects [Including interface]	99.5% compliance with purchase/product instructions	95%-99%	20% of the total PO amount of this quarter will be deducted;
2				90%-95%	50% of the total PO amount of this quarter will be deducted;
3				80%-90%	100% of the total PO amount of this quarter will be deducted;
4	Performance index	response time	2 seconds	2 seconds to 3 seconds	20% of the total PO amount of this quarter will be deducted;
5				More than 3 seconds	50% of the total PO amount of this quarter will be deducted;
6		Resource usage (CPU, memory)	<80%	80%-85%	20% of the total PO amount of this quarter will be deducted;
7				85%-95%	50% of the total PO amount of this quarter will be deducted;
8				More than 95%	100% of the total PO amount of this quarter will be deducted;
9		The stability of [Uptime/(uptime + failure time)]	The stability of > 99.5%	The stability of <99.5%	20% of the total PO amount of this quarter will be deducted;
10				99%<The stability of<=99.5%	50% of the total PO amount of this quarter will be deducted;
11				The stability of<99%	100% of the total PO amount of this quarter will be deducted;

N O.	Index type	Index name	target value	index value	penalty
12		Robustness, [Average trouble-free time/day]	Continuous normal operation for 12 hours	10-12 hours	20% of the total PO amount of this quarter will be deducted;
13				Less than 10 hours	50% of the total PO amount of this quarter will be deducted;